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NONWOVEN FABRIC

Abstract

A nonwoven fabric including fibers, containing a component A below, a component B below, inside the fibers, wherein the nonwoven fabric includes 50 mass % or more and 95 mass % or less of the component A and 5 mass % or more and 50 mass % or less of the component B with respect to a mass of the whole nonwoven fabric, the fibers included in the nonwoven fabric have a number average fiber diameter of 4 μm or less, the nonwoven fabric has a sheet width of 50 mm or more, the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than 150° C.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation of PCT International Application No. PCT/JP2023/027985, filed on Jul. 31, 2023, which claims priority under 35 U.S.C. § 119 (a) to Japanese Patent Application No. 2022-189675, filed in Japan on Nov. 28, 2022. Each of the above applications is hereby expressly incorporated by reference, in its entirety, into the present application.

SUMMARY

[0002] According to one or more aspects, a method of producing fibers that spins a thermoplastic resin composition including 50 mass % or more and 95 mass % or less of a component A below and 5 mass % or more and 50 mass % or less of a component B'

below with respect to a mass of the whole thermoplastic resin composition with a fiber diameter of 4 μm or less can be implemented, and the method can comprise: a process of heating and melting the thermoplastic resin composition (I); and a process of discharging the thermoplastic resin composition from a nozzle (II), wherein the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B' is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than a process temperature.

[0003] Optionally, the component B can include a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and having a melting point lower than 150° C.

[0004] According to another aspect, a method of producing fibers that spins a thermoplastic resin composition including a component A and a component B' below can be implemented.

[0005] Optionally, the component A can include a thermoplastic resin having a solidification point of 100° C. or less.

[0006] Optionally, the component B' can include a compound having a solidification point higher than a solidification point of the thermoplastic resin and a melting point lower than a process temperature.

[0007] Optionally, the component A can have a contained amount of 50 mass % or more and 95 mass % or less with respect to a mass of the whole thermoplastic resin composition.

[0008] Optionally, the component B' can have a contained amount of 5 mass % or more and 50 mass % or less with respect to the mass of the whole thermoplastic resin composition.

[0009] Optionally, the thermoplastic resin composition can undergo a process of heating and melting the thermoplastic resin composition (I) and a process of discharging the thermoplastic resin composition from a nozzle (II).

[0010] The thermoplastic resin composition may be spun with a fiber diameter of 4 μm or less.

[0011] According to another aspect, a nonwoven fabric including fibers can be provided or implemented. The nonwoven fabric can comprise: a component A below; and a component B below, inside the fibers, wherein the nonwoven fabric includes 50 mass % or more and 95 mass % or less of the component A and 5 mass % or more and 50 mass % or less of the component B with respect to a mass of the whole nonwoven fabric, the fibers included in the nonwoven fabric have a number average fiber diameter of 4 μm or less, the nonwoven fabric has a sheet width of 50 mm or more, the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than 150° C.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. 1 is a perspective cross-sectional view schematically illustrating an example of a fiber constituting a nonwoven fabric according to one or more embodiments of the present disclosure.

[0013] FIG. 2(A) is a graph showing respective results of DSC (temperature decreases) of a thermoplastic resin of a component A, a compound of a component B, and a thermoplastic resin composition fabricated by mixing the thermoplastic resin of the component A and the compound of the component B in Example 1.

[0014] FIG. 2(B) is a graph showing relations between their temperatures and viscosities.

[0015] FIG. 2(C) is a graph showing relations between their temperatures and elastic moduli.

[0016] FIG. 3(A) is a photograph substituted for a drawing showing a fiber state of a nonwoven fabric sample obtained in Example 1.

[0017] FIG. 3(B) is a graph showing a number distribution of the fiber diameters.

[0018] FIG. 3(C) is a graph showing a volumetric distribution of the fiber diameters.

[0019] FIG. 4 is a photograph substituted for a drawing showing a state of a sample obtained in Comparative Example 1.

DESCRIPTION OF EMBODIMENTS

[0020] The present disclosure relates to providing a wide nonwoven fabric including ultrafine fibers and a method of producing the ultrafine fibers constituting the nonwoven fabric.

[0021] When ultrafine fibers are produced using the aforementioned low-melting-point thermoplastic resin, the conventional techniques have difficulty in productivity, and the reality is that the industrialization has not been achieved. Specifically, in order to thin a fiber, an electrospinning method (a melting method, a solution method) and a melt-blown method generally have restrictions requiring a reduced discharge amount from a nozzle and slow spinning.

[0022] In addition, the low-melting-point thermoplastic resin is likely to maintain its fluidity even after being discharged from the nozzle in a molten state or a solution state. Therefore, some may exhibit liquid-like behavior resulting in an extension shortage, or some may induce fusion without being sufficiently cooled resulting in an agglomerate. In view of this, ball-shaped products (shots) and fiber breakage (fly-shaped products) are likely to occur, and thus, stable fiberization with good productivity is difficult.

Therefore, it may be necessary or desirable to further reduce the discharge amount. For example, there has been required a low speed of approximately 1/100 to 1/1000 of a spinning speed generally desired in industrial production on a producing line so as to set the discharge amount per nozzle to approximately 0.001 g/minute.Math.nozzle.

[0023] An extreme reduction of the discharge amount has been thus necessary, and therefore, a nonwoven fabric with ultrafine fibers using the low-melting-point thermoplastic resin has been difficult to industrially produce. Therefore, a wide nonwoven fabric having a desired width with ultrafine fibers (for example, a nonwoven fabric with a good quality having a sufficient width required for a practical product, such as a filter) has been difficult to efficiently produce in quantity on a conventional producing line, and therefore, the development of a wide nonwoven fabric constituted of ultrafine fibers has been desired from the aspect of industrial production.

[0024] A nonwoven fabric of one or more embodiments of the present disclosure can be or include a nonwoven fabric (e.g., a wide nonwoven fabric) including ultrafine fibers. With a method of producing fibers of one or more embodiments of the present

disclosure, the ultrafine fibers constituting the nonwoven fabric can be efficiently produced in quantity.

[0025] The following describes the nonwoven fabric and the method of producing fibers constituting the nonwoven fabric of one or more embodiments of the present disclosure.

[0026] The nonwoven fabric of one or more embodiments of the present disclosure can include a thermoplastic resin (hereinafter referred to as a component A) having a solidification point of 100° C. or less and a compound (hereinafter referred to as a component B) having a solidification point higher than the solidification point of the thermoplastic resin and a melting point of lower than 150° C. inside the fiber.

[0027] The nonwoven fabric of one or more embodiments of the present disclosure can include 50 mass % or more and 95 mass % or less of the component A and 5 mass % or more and 50 mass % or less of the component B with respect to a mass of the whole nonwoven fabric of one or more embodiments of the present disclosure. The above-described respective content percentages of the component A and the component B mentioned here can mean percentages when the mass of the whole nonwoven fabric of one or more embodiments of the present disclosure is 100 mass %.

(Method for Measuring Solidification Point)

[0028] The “solidification point” may also be referred to as a solidification temperature, and can mean a peak temperature of an exothermic peak that first appears when the temperature of a sample is increased, and after the sample melts, is decreased at 5° C./minute in differential scanning calorimetry (DSC). This measurement may be specifically performed as follows. A nonwoven fabric is washed with a solvent that does not dissolve fibers, such as water, and dried. Thereafter, a segment is cut out to have a mass of 1 mg, from which measurement target constituent components are extracted. When the nonwoven fabric is incorporated in a product, only a corresponding part may be separated or cut out to take the nonwoven fabric out, and then, the extraction is performed. The extracted constituent component is sealed in a sample pan made of aluminum and is heated, and the temperature is increased at 5° C./minute. After the heating temperature reaches 200° C. by the temperature increase, the temperature is decreased at 5° C./minute within 600 seconds. The measurement can be terminated at the point when the temperature reaches 0° C.

[0029] The above-described peak temperature can mean a temperature at which the heated component starts to coagulate by the temperature decrease from a molten state. The “molten state” can be a state in which the above-described component flows when an external force is applied, and, for example, means a state of being heated to the melting point or more of the target component. The “coagulation” can mean crystallization, or when the crystallization is not observed, means glass transition.

(Method for Measuring Content Percentages of Component a and Component B)

[0030] The constituent components extracted by the extraction method of each constituent component described in (Method For measuring solidification point) described above are dissolved in solvents in which the respective constituent components are soluble and that are deuterated, and each constituent component is identified using proton NMR. In view of this, the constituent components corresponding to the components A and B are identified.

[0031] Next, the component is extracted from a fiber aggregation with a solvent that can dissolve the identified component A or B, and thus, a content percentage thereof is obtained.

[0032] For example, the fiber aggregation is immersed in an organic solvent that can dissolve the component B for 24 hours to extract the component B. Fibers are extracted from the organic solvent, and are dried for 24 hours at 40° C. under a reduced pressure of -0.04 MPa. Thereafter, measuring a fiber weight after the drying allows measurement of mass % of the component B.

[00001]

Component *B* content percentage (mass %) = 100 - (fiber mass after drying under reduced pressure / initial fiber mass) × 100

[0033] For each of the constituent components identified above, the measurement target fiber is measured by Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS) to detect whether each of the constituent components is detected inside a fiber shape or not. When the constituent components corresponding to the components A and B are detected inside the fiber shape, it is determined that they are included inside the fiber.

[0034] The nonwoven fabric contains 50 mass % or more and 95 mass % or less of the component A with respect to the mass of the whole nonwoven fabric, and the component A serves as a main base of constituent fibers of the nonwoven fabric of one or more embodiments of the present disclosure. Such a thermoplastic resin has a solidification point of 100° C. or less, and thus, the constituent fibers containing this as the main base have viscosity and elasticity reduction at a relatively low temperature. This can enable the nonwoven fabric of one or more embodiments of the present disclosure to have a low lamination temperature by embossing or the like, making the nonwoven fabric of one or more embodiments of the present disclosure have high workability. This can also or alternatively facilitate spinning in a producing method described below, and since heat bonding or the like can be possible at a low temperature, energy consumption and investment in plant and equipment can be reduced.

[0035] From the above-described aspects, the thermoplastic resin included in the component A has a solidification point of preferably 75° C. or less, and more preferably 60° C. or less.

[0036] For instance, the solidification point of the thermoplastic resin included in the component A can be 0° C. or more.

[0037] The thermoplastic resin included in the component A has a melting point of preferably 150° C. or less, more preferably 120° C. or less, and further preferably 90° C. or less. This can enable the lamination temperature by embossing or the like to be low, making the nonwoven fabric of the present invention have high workability.

[0038] The melting point is preferably 50° C. or more. This can increase shape stability in a high temperature environment, such as in summer.

[0039] The component A has a content percentage with respect to the mass of the whole nonwoven fabric of preferably 60 mass % or more, more preferably 65 mass % or more, and further preferably 70 mass % or more, for instance, from the aspect of further enhancing the above-described effects.

[0040] The component A has a content percentage with respect to the mass of the whole nonwoven fabric of preferably 95 mass % or less, more preferably less than 90 mass %, and further preferably 85 mass % or less, for instance, from the aspect of further enhancing effects of the component B described below.

[0041] The component B has a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A, the nonwoven fabric contains 5 mass % or more and 50 mass % or less of the component B with respect to the mass of the whole nonwoven fabric, and thus, a solidification point of

the constituent fibers of the nonwoven fabric of one or more embodiments of the present disclosure can be appropriately increased from the solidification point of the component A alone. In the producing method described below, a compound included in the component B may be solidified earlier than the thermoplastic resin included in the component A during a process in which a melt of a thermoplastic resin composition (component A+component B) discharged from the nozzle is extended and cooled.

[0042] In the producing method described below, the whole thermoplastic resin composition has a reference temperature at which the viscosity and the elastic modulus are rapidly increased from the molten state, and the reference temperature is more appropriately increased than the case of the component A alone due to the appropriate increase in the solidification point. In view of this, the timing of the solidification after discharging can be improved. While the component A as the main base of the fibers is being extended in the molten state, the component B supplements a strength of an ultra-thin diameter of 4 μm or less to improve the extension, thus enabling satisfactory fiberization. Since the melting point of the component B is lower than 150° C., the component B may be present in the molten state in the melt discharged from the nozzle under an environment at a process temperature in the producing method described below. In view of this, in the producing method described below, mix meltability of the component B and the component A can be enhanced, the generation of a resin agglomerate mainly caused by the component B can be reduced, and therefore, clogging of the nozzle and fiber breakage can be less likely to occur.

[0043] As a result, the nonwoven fabric of one or more embodiments of the present disclosure can have reduced defects, such as a hole. In addition, the generation of the ball-shaped products can be reduced, the fiber diameter can become thinner, and uniformity thereof can be enhanced. The nonwoven fabric of one or more embodiments of the present disclosure can be regarded as of good quality with a satisfactory texture.

[0044] From the aspect of further improving the above-described effects, for instance, a difference between the solidification point of the compound included in the component B and the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A is preferably 5° C. or more.

[0045] From the aspect of reducing extra energy consumption, for instance, the difference is preferably 100° C. or less, and more preferably 60° C. or less.

[0046] The melting point of the compound included in the component B is preferably less than 150° C., more preferably 135° C. or less, and further preferably 120° C. or less, for instance, from the aspect of further improving the above-described effects.

[0047] The melting point of the compound included in the component B is preferably 30° C. or more, more preferably 40° C. or more, and further preferably 50° C. or more, for instance, from the aspect of facilitating handling in raw material supply.

[0048] From the aspect of further enhancing the miscibility between the component A and the component B in the producing method described below, for instance, when focusing on a combination where a difference between a melting point of a thermoplastic resin included in the component A and a melting point of a compound included in the component B becomes the largest, the difference has a value of more than 0° C. and preferably 100° C. or less, more preferably 60° C. or less, and further preferably 50° C. or less.

[0049] Falling within this temperature range can enable kneading without excessively heating one material and kneading for a sufficient period of time for mixing.

[0050] The content percentage of the component B with respect to the mass of the whole nonwoven fabric is preferably 10 mass % or more, and more preferably more than 10 mass %, for instance, from the aspect of further enhancing the above-described effects.

[0051] The content percentage of the component B with respect to the mass of the whole nonwoven fabric is preferably 40 mass % or less, more preferably 35 mass % or less, and further preferably 30 mass % or less. Falling at or below the above-described upper limit enables sufficient mixing with the component A by a melt-kneading machine, such as an extruder, and therefore, the above-described effects can be further enhanced.

[0052] The nonwoven fabric of one or more embodiments of the present disclosure can include the component A and the component B described above as the constituent components of the constituent fibers, and can be produced as a satisfactory nonwoven fabric by the producing method described below. As a result, the nonwoven fabric of one or more embodiments of the present disclosure can have a wide sheet width with a ultra-thin number average fiber diameter of 4 μm or less.

[0053] The number average fiber diameter can be a fiber diameter measured by a method below, and can indicate an average fiber diameter of the whole constituent fibers of the nonwoven fabric of the present invention. Such a number average fiber diameter is preferably 3 μm or less, and more preferably 2 μm or less, for instance, from the aspect of providing a softer and smoother touch of the nonwoven fabric.

[0054] The number average fiber diameter is preferably 0.1 μm or more, and more preferably 0.2 μm or more, for instance, from the aspect of maintaining a fiber strength.

(Method for Measuring Number Average Fiber Diameter of Constituent Fibers of Nonwoven Fabric of Present Invention)

[0055] A hundred fibers from which defects, such as lumps of the constituent fibers, intersecting portions of the constituent fibers, and polymer droplets, are eliminated are randomly selected from a two-dimensional image by scanning electron microscope observation, and a width perpendicular to a longitudinal direction (a fiber length direction) of every single fiber can be measured. The sum of these values can be divided by the number of the measured fibers to obtain a number average fiber diameter of the measurement target nonwoven fabric. When a cross-sectional surface perpendicular to the longitudinal direction of the constituent fiber is not a circle, the above-described fiber diameter can be converted into an equivalent circle diameter.

[0056] In the nonwoven fabric of one or more embodiments of the present disclosure, a median fiber diameter of the constituent fibers is preferably 2 μm or less, more preferably 1.5 μm or less, and further preferably 1 μm or less, for instance, from the aspect of further improving softness and smoothness of touch.

[0057] From the aspect of maintaining the fiber strength, for instance, the median fiber diameter of the constituent fibers is preferably 0.1 μm or more, and more preferably 0.2 μm or more.

[0058] The median fiber diameter mentioned here can mean a fiber diameter having a cumulative frequency of 50% (a median) of the whole in a frequency distribution (histogram) of the fiber diameters obtained by the measuring method of the aforementioned number average fiber diameter.

[0059] In the nonwoven fabric of one or more embodiments of the present disclosure, satisfying the requirement of the median fiber diameter in addition to the requirement of the aforementioned number average fiber diameter can increase the number of thinner fibers, thereby improving wipeability and adhesion to the skin.

[0060] In the nonwoven fabric of one or more embodiments of the present disclosure, the aforementioned “seat width” can mean a length of the longest line segment among line segments connecting outer lines passing through the center of a plane of the nonwoven fabric of the present invention. For example, when the plane is a circle, it is a diameter thereof, and when the plane is an oval, it is a major axis. When the plane is a quadrilateral, it is a length of the longest line segment among straight lines connecting two opposing sides passing through the center as described above. When the plane is a square, it is equivalent to a length of one side, and when the plane is a rectangle, it is equivalent to a length of a long side. When the plane is a planar shape not included in the above-described shapes, such as a polygon, an equivalent circle diameter calculated from a size is regarded as the sheet width. However, in a case of a continuous sheet, the center of the plane may not be determined, and therefore, a length along a width direction perpendicular to the longitudinal direction can be the sheet width for convenience.

[0061] A nonwoven fabric fabricated from fibers with similar fiber diameters by a conventional method has had a significantly small discharge amount from the nozzle, and therefore, required a long time for fabricating a sheet with a wide sheet width, thereby not being realistic. In contrast to this, since the constituent fibers of the nonwoven fabric of one or more embodiments of the present disclosure can include (or consist of) the component A and the component B mentioned above, the aforementioned ultra-thin fiber diameter can be efficiently and quickly formed in the producing method described below, and the nonwoven fabric of one or more embodiments of the present disclosure can have a larger size with enhanced uniformity of the ultra-thin fiber diameters. The sheet width of the nonwoven fabric of one or more embodiments of the present disclosure is preferably 50 mm or more, more preferably 100 mm or more, further preferably 150 mm or more, further preferably 200 mm or more, and further preferably 300 mm or more.

[0062] The longer the sheet width is, the more preferable it may be, and the upper limit is realistically 3 000 mm or less, for instance, from the aspect of reducing the increase in the width of a producing apparatus.

[0063] From the aspect of a dimension required for an actual product, for instance, the “sheet width”/“length perpendicular to the sheet width” is preferably 8 or less, more preferably 4 or less, further preferably 2 or less, and still more preferably 1.5 or less.

[0064] Similarly, from the aspect of the dimension required for an actual product, for instance, the “sheet width”/“length perpendicular to the sheet width” can be 1 or more.

[0065] In the nonwoven fabric of one or more embodiments of the present disclosure, the component A can include various kinds of thermoplastic resins having solidification points of 100° C. or less.

[0066] For example, the component A is allowed to include one or two or more selected from polyolefin resins, polyester resins, aliphatic polyamide resins, vinyl-based polymer resins, acrylic-based polymer resins, polyvinyl acetate, polyvinyl acetate-ethylene copolymers, and polyether resins.

[0067] The polyolefin resin is allowed to include one or two or more selected from polyethylene, polypropylene, and ethylene- α -olefin copolymers.

[0068] The polyester resin is allowed to include one or two or more selected from aliphatic polyesters, semi-aromatic polyesters, polybutylene terephthalate, polylactic acid, polyhydroxyalkanoates, and liquid crystal polymers.

[0069] The aliphatic polyester is allowed to include one or two or more selected from polycaprolactone, polybutylene succinate, polybutylene succinate adipate, polyglycolic acid, and polydioxanone.

[0070] The semi-aromatic polyester is allowed to include one or two or more selected from polyethylene terephthalate and polybutylene terephthalate.

[0071] The aliphatic polyamide resin is allowed to include one or two or more selected from nylon 6 and nylon 66.

[0072] The vinyl-based polymer resin is allowed to include one or two or more selected from polyvinyl chloride, polyvinylidene chloride, and polystyrene.

[0073] The acrylic-based polymer resin is allowed to include one or two or more selected from polyacrylic acid, polyacrylic acid esters, polymethacrylic acid, and polymethacrylic acid esters.

[0074] The polyether resin is allowed to include polyethylene oxide.

[0075] Among these, the component A preferably includes one or two or more selected from polyester resins and polyether resins from the aspect of enhancing the miscibility with the component B.

[0076] The aliphatic polyester resin is preferably biodegradable. This can enable reducing an environmental impact in a case where the constituent fibers of the nonwoven fabric of the present invention flow out to the environment (for example, in a case where the nonwoven fabric of one or more embodiments of the present disclosure can be used as a cosmetic material, and the fibers flow when it is washed for reuse or the like). Note that “biodegradable” mentioned here can mean an aliphatic polyester resin having a biodegradation degree of polyester of 30% or more measured in compliance with Japanese Industrial Standard K 6953-1.

[0077] The biodegradable aliphatic polyester resin preferably includes one or two or more selected from polycaprolactone (hereinafter also referred to as PCL), polybutylene succinate, polybutylene succinate adipate, polydioxanone, and polyglycolic acid.

[0078] The biodegradable polyether resin preferably includes polyethylene oxide.

[0079] From the aspect of improving workability and biodegradability, for instance, it is more preferred to include PCL among the polyester resins.

[0080] Any one of the above preferably satisfies the requirement of a mass average molecular weight described below while having a solidification point of 100° C. or less.

[0081] As the main base of the fibers, the thermoplastic resin included in the component A has a mass average molecular weight of preferably 200 000 g/mol or less, more preferably 150 000 g/mol or less, and further preferably 100 000 g/mol or less from the aspect of obtaining the nonwoven fabric of one or more embodiments of the present disclosure having a larger size with a thin fiber diameter mentioned above.

[0082] The mass average molecular weight of the thermoplastic resin included in the component A is preferably 5 000 g/mol or more, and more preferably 10 000 g/mol or more, for instance, from the aspect of successful extension in a molten state during spinning.

(Method for Measuring Mass Average Molecular Weight)

[0083] Based on the components A, B identified in (Method for measuring content percentages of Component A and Component B) described above, a nonwoven fabric piece of the measurement target is immersed in a solvent that dissolves only any one of the component A and the component B, the solvent and the residue are separated, and thereafter, the component A or the component B is

separated by drying.

[0084] The molecular weight of the component B is obtainable from the measurement in (Method for measuring content percentages of Component A and Component B) described above.

[0085] Using gel permeation chromatography, the separated component A is measured in accordance with the following conditions with a polystyrene conversion mass average molecular weight. Polystyrene samples (for example, monodispersed polystyrene manufactured by Tosoh Corporation (model number: F450, F288, F128, F80, F40, F20, F10, F4, F1, A5000, A2500, A1000, A500, and A300)) whose mass average molecular weights are already known and whose mass average molecular weights are different from one another are used as polystyrene standard samples to preliminarily make molecular weight calibration curves, and the measurement is taken by comparing the calibration curves with the results of the measurement samples.

<Gel Permeation Chromatography Conditions>

[0086] Measurement device: HLC-8220GPC (manufactured by Tosoh Corporation) [0087] Column: GMHHR-H+GMHHR-H (manufactured by Tosoh Corporation) [0088] Eluent: 1 mmol FARMIN DM20 (manufactured by Kao Corporation)/CHCl₃ [0089] Eluent flow rate: 1.0 mL/min [0090] Column temperature: 40° C. [0091] Detector: RI [0092] Sample concentration: 0.1 volume % (chloroform solution) [0093] Sample injection amount: 100 mL

[0094] In the nonwoven fabric of one or more embodiments of the present disclosure, the compound included in the component B has a mass average molecular weight of preferably 1 000 g/mol or less. A polymer can be avoided, and thus, the miscibility with the thermoplastic resin of the component A can be enhanced.

[0095] For instance, the compound included in the component B can be 100 g/mol or more.

[0096] For the compound included in the component B, those having a solidification point higher than the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A and having a melting point lower than 150° C. are variously employable.

[0097] For example, the component B is allowed to include one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyhydric alcohol organic acid ester compounds, waxes, sphingolipids, alkyl ammonium salts, and fatty acid metal salts.

[0098] The fatty acid is allowed to include one or two or more selected from, for example, stearic acid, behenic acid, myristic acid, and palmitic acid.

[0099] The higher alcohol is allowed to include one or two or more selected from, for example, cetanol, myristyl alcohol, and stearyl alcohol.

[0100] The fatty acid amide compound is allowed to include one or two or more selected from, for example, stearic acid monoamide, oleic acid amide, erucic acid amide, and ethylenebisstearamide.

[0101] The polyhydric alcohol organic acid ester compound is allowed to include one or two or more selected from, for example, ester compounds of polyhydric alcohol and fatty acid, ester compounds of polyhydric alcohol and polycarboxylic acid, and ester compounds of polyhydric alcohol and hydroxy acid.

[0102] This ester compound may be a monoester, may be a diester, or may be a triester.

[0103] The polyhydric alcohol organic acid ester compound is allowed to include one or two or more selected from, for example, glycerin fatty acid ester compounds, polyglyceryl fatty acid ester compounds, sucrose fatty acid ester compounds, and sorbitan fatty acid ester compounds.

[0104] The glycerin fatty acid ester compound is allowed to include one or two or more selected from, for example, glyceryl behenate, glyceryl stearate, and 12-glyceryl hydroxystearate.

[0105] The polyglyceryl fatty acid ester compound is allowed to include one or two or more selected from, for example, polyglyceryl stearate and polyglyceryl behenate.

[0106] The sucrose fatty acid ester compound is allowed to include one or two or more selected from, for example, sucrose stearic acid ester compounds and sucrose glyceryl behenate.

[0107] The sorbitan fatty acid ester compound is allowed to include one or two or more selected from, for example, sorbitan distearate, sorbitan monostearate, and sorbitan tribehenate.

[0108] The wax is allowed to include one or two or more selected from, for example, various kinds of vegetable waxes, such as paraffin waxes and carnauba waxes.

[0109] The sphingolipid is allowed to include one or two or more selected from, for example, ceramides, glycosphingolipids, and sphingophospholipids.

[0110] The alkyl ammonium salt is allowed to include one or two or more selected from, for example, distearyldimonium chloride.

[0111] The fatty acid metal salt is allowed to include one or two or more selected from, for example, zinc stearate, magnesium stearate, and calcium stearate.

[0112] In the specific examples of the above-described compounds, those satisfying the requirements for the component B preferably satisfy the aforementioned mass average molecular weight requirement.

[0113] From the aspect of mixing with the component A, for instance, the component B preferably includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyhydric alcohol organic acid ester compounds, waxes, and fatty acid metal salts.

[0114] From the aspect of further enhancing the miscibility between the component B and the component A, for instance, it is preferred that the component A includes an aliphatic polyester resin with polarity and the component B includes one or two or more selected from fatty acids, higher alcohols, and ester compounds of polyhydric alcohol and fatty acid.

[0115] In this regard, the ester compound can have a skeleton having a single bond between carbon atoms, and it is more preferred that the skeleton is a straight chain.

[0116] Among these, the component B more preferably includes one or two or more selected from stearic acid, behenic acid, myristyl alcohol, and glyceryl behenate. Selecting these components enables obtainment of the nonwoven fabric with thin fiber diameters.

[0117] Among the compounds included in the component B, a compound having a salt structure, in particular, a metal salt structure is preferably 5 mass % or less, more preferably 3 mass % or less, and further preferably 1 mass % or less. This can enable preventing

an electric charge from escaping through the component B in the fibers during an electrostatic spinning process.

[0118] Among the compounds included in the component B, the compound having a salt structure, in particular, a metal salt structure is preferably more than 0 mass %, and more preferably 0.5 mass % or more. This can enable effectively performing the electrostatic spinning process.

[0119] The “salt structure” mentioned here can mean a structure having positive electric charge portions and negative electric charge portions, in which they are ionized during melting. For example, the examples can include a structure having positive electric charge portions and negative electric charge portions in one molecule (also referred to as a zwitter ion or an inner salt), and the one molecule is in a neutral state. The salt structure may form a salt between a plurality of molecules of cationic compounds having the positive electric charge portions in one molecule and anionic compounds having the negative electric charge portions in one molecule.

[0120] In such a fiber of one or more embodiments of the present disclosure, it is preferred that the component A constitutes a core portion layer of the fiber and extends in the longitudinal direction (a fiber length direction) of the fiber, and the component B is arranged on the fiber surface (that is, a surface of the thermoplastic fiber). In this case, while the component B can be present inside the fiber, a part thereof is shown on the fiber surface side. Some parts of the component B inside the fiber may be mixed with the component A.

[0121] It is preferred that a part of the component B covers a peripheral surface of a core portion layer 2 of the component A as a skin layer 3 as a constituent fiber 1 illustrated in FIG. 1. In this regard, an interface between the component concentrations of the skin layer 3 of the component B and the core portion layer 2 of the component A does not necessarily have to be clear, and is preferably blurry. The skin layer 3 of the component B may cover the entire fiber surface, or may partially cover the fiber surface. When the fiber surface is partially covered, the arrangement may be a sea-island structure including a region without the skin layer 3 of the component B in a region with the skin layer 3 of the component B, or the arrangement may have the region with the skin layer 3 of the component B and the region without the skin layer 3 of the component B separated.

[0122] The core portion layer 2 and the skin layer 3 can be formed as follows in the producing method described below.

[0123] When the component A and the component B having different solidification points are spun, one with a higher solidification point is solidified earlier during the spinning and therefore stabilized. As one which shows stability at an interface between air and the molten resin emerges there, the one that is easily solidified is likely to be formed at the side of the air. Therefore, while the component A and the component B can be present inside the fibers, the core portion layer 2 and the skin layer 3 can be formed due to the difference in solidification point.

[0124] Next, one or more embodiments of the method of producing the fibers constituting the nonwoven fabric will be described.

[0125] The method of producing the fibers of the embodiment can use a thermoplastic resin composition. The thermoplastic resin composition preferably includes 50 mass % or more and 95 mass % or less of a thermoplastic resin (hereinafter referred to as the component A) having a solidification point of 100° C. or less, and 5 mass % or more and 50 mass % or less of a compound (hereinafter referred to as a component B') having a solidification point higher than the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than a process temperature with respect to a mass of the whole thermoplastic resin composition. The thermoplastic resin composition preferably undergoes a process of heating and melting the thermoplastic resin composition (I) and a process of discharging the thermoplastic resin composition from a nozzle (II). Thus, the thermoplastic resin composition is preferably spun into fibers with a fiber diameter of 4 μm or less. The component B' can have properties similar to those of the component B described in the aforementioned nonwoven fabric of one or more embodiments of the present disclosure, and can include the compounds described as specific examples of the component B.

[0126] The “process temperature” can mean the temperature of heat applied in the producing method of one or more embodiments of the present disclosure, and can mean (e.g., mainly) the heating temperature for heating and melting the thermoplastic resin composition in the process (I). In addition, the “process temperature” can mean the temperature of a heating fluid described below sprayed in the process (II). This process temperature is appropriately set corresponding to a melting point of the thermoplastic resin having the highest melting point among the thermoplastic resins included in the component A as a main base of the fibers. The process temperature is set higher than the melting point of the thermoplastic resin having the highest melting point among the thermoplastic resins included in the component A in consideration of the melting point of the component B' as well (for example, the melting point+20° C. or more and 200° C. or less). For example, the process temperature is 150° C.

[0127] The component A has a content percentage with respect to the whole mass of the thermoplastic resin composition of preferably 60 mass % or more, more preferably 65 mass % or more, and further preferably 70 mass % or more.

[0128] The component A has a content percentage with respect to the whole mass of the thermoplastic resin composition of preferably 95 mass % or less, more preferably less than 90 mass %, and further preferably 85 mass % or less.

[0129] The component B' has a content percentage with respect to the whole mass of the thermoplastic resin composition of preferably 10 mass % or more, and more preferably more than 10 mass %.

[0130] The component B' has a content percentage with respect to the whole mass of the thermoplastic resin composition of preferably 40 mass % or less, more preferably 35 mass % or less, and further preferably 30 mass % or less.

[0131] Thus, the effects of the component A and the component B' in the producing method described in the description regarding the aforementioned nonwoven fabric can be enhanced.

[0132] In the process (I), for example, via a hopper, the component A and the component B' are added in a housing connected to the hopper. These component A and component B' are heated and melted in the housing, a melt of the thermoplastic resin composition (hereinafter also simply referred to as a resin mix melt) is fabricated. This resin mix melt is pressed out toward a discharging nozzle by the rotation of a screw, and is supplied to a discharge port of a nozzle head portion. The nozzle in this case may be one or may be plural.

[0133] Next, the supplied resin mix melt is discharged from the nozzle and spinning is performed in the process (II). The discharged resin mix melt is extended and cooled to be solidified and becomes a fiber as it moves away from the discharge port of the nozzle head portion. At this time, a hole diameter of the discharge port of the nozzle head portion is appropriately set, and thus, the aforementioned ultrafine fibers having a fiber diameter of preferably 4 μm or less can be spun. As mentioned above, since the resin mix melt of the component A and the component B is spun, ball-shaped products (shots) and fiber breakage (fly-shaped products) are

less likely to be generated even though the spinning is performed at a high discharge speed (for example, 0.5 g/minute to 5 g/minute) at which the spinning is industrially performed on an actual producing line, not at a low discharge speed of a conventional specimen producing level. This enables stably fiberizing ultrafine fibers with a number average fiber diameter of 4 μm or less in a uniform and efficient manner.

[0134] In the process (II), the discharge speed of the melt of the thermoplastic resin composition from the nozzle is preferably 0.1 g/minute.Math.nozzle or more, more preferably 0.2 g/minute.Math.nozzle or more, and further preferably 0.5 g/minute.Math.nozzle or more.

[0135] The discharge speed of the melt of the thermoplastic resin composition from the nozzle is preferably 20 g/minute.Math.nozzle or less, and more preferably 10 g/minute.Math.nozzle or less from the aspect of thinning the diameter of the fiber to be spun.

[0136] The viscosity of the resin mix melt when it is discharged from the nozzle is preferably 1 Pa.Math.s or more, more preferably 2 Pa.Math.s or more, and further preferably 5 Pa.Math.s or more, for instance, from the aspect of further enhancing uniform spinnability with a ultrafine fiber diameter.

[0137] The viscosity of the resin mix melt when it is discharged from the nozzle is preferably 40 Pa.Math.s or less, more preferably 20 Pa.Math.s or less, and further preferably 15 Pa.Math.s or less, for instance, from the aspect of reducing the viscosity to easily thin the fibers.

(Method for Measuring Viscosity of Resin Mix Melt)

[0138] The melt viscosity can be measured using a rotational rheometer. Specifically, the measurement is taken using MCR305 manufactured by Anton Paar GmbH. The viscosity measurement is taken at a shear rate of 0.1 s^{sup}.-1 using a q 50 mm parallel plate as a measuring tool. The measurement temperature is set to a temperature corresponding to the spinning conditions. A sample is set on the plate, a clearance is set to 1 mm after the resin melts, and a portion protruding from the q 50 mm parallel plate is trimmed. Thereafter, the measurement is held until the sample reaches the measurement temperature, and then, the measurement is started. For obtainment of a viscosity value, a value 100 seconds after the start of rotation is used as a measured value.

[0139] The method of producing the fibers of one or more embodiments can preferably perform a heating fluid spraying process in the process (II). This spraying is performed on the resin mix melt in a state before the resin mix melt discharged from the nozzle completely solidifies. The heat of the sprayed heating fluid can enable more active extension of the discharged resin mix melt, thus enabling further ultra-thin fibers to be formed. The spraying of the heating fluid may be performed along a discharging direction of the resin mix melt, or may be performed in a direction intersecting with the discharging direction.

[0140] The heating fluid preferably has a temperature higher than the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A, for instance, from the aspect of making the above-described extension more effective.

[0141] Specifically, a difference between the temperature of the heating fluid and the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A is preferably 30° C. or more, more preferably 40° C. or more, and further preferably 50° C. or more.

[0142] A difference between the temperature of the heating fluid and the solidification point of the thermoplastic resin of the component A is preferably 200° C. or less, more preferably 150° C. or less, and further preferably 130° C. or less, for instance, from the aspect of inhibiting decomposition of the resin.

[0143] The method of producing the fibers of one or more embodiments can perform an electrostatic spinning process in the process (II). This electrostatic spinning process may be performed together with the heating fluid spraying process described above, or may be performed instead of the heating fluid spraying process.

[0144] The electrostatic spinning process can be or include a process also referred to as an electrospinning method, and is a process of spinning by directly or indirectly charging the nozzle from which the resin is discharged to apply an electric charge to the resin. This can enable more active extension, thus enabling a further ultra-thin fibers to be formed. For example, a charged electrode and a high-voltage generator connected to the charged electrode are disposed at a corresponding position separated from the nozzle. This configuration can allow application of a high voltage between a nozzle head portion and the charged electrode to form an electric field therebetween, thus enabling the resin mix melt discharged from the nozzle head portion to be charged. The charged electrode is preferably configured of a conductive material, such as metal, or covered with a dielectric material.

[0145] In the method of producing the fibers of one or more embodiments of the present disclosure, in addition to the component A and the component B', another agent may be further included as long as the effects of one or more embodiments of the present disclosure are not impaired. For example, examples of the other agent include charge control agents, lubricants, antistatic agents, hydrophilizing agents, surfactants, plasticizers, and the like from the aspect of increasing the above-described electric charge amount. Other than those, for instance, antioxidants, neutralizers, light stabilizers, UV absorbers, and the like may be included.

[0146] Through a process of collecting the fibers thus obtained by the method of producing the fibers of one or more embodiments and forming the fibers into a sheet shape, the aforementioned nonwoven fabric of one or more embodiments of the present disclosure can be preferably produced. For example, the resin mix melt discharged from the nozzle head portion is collected by a collection portion and deposited into a sheet shape while being cooled and extended, and thus, can be formed into a nonwoven fabric. The collection portion preferably includes a collection electrode and a high-voltage generator connected to the collection electrode from the aspect of enhancing capture efficiency. The collection electrode and the high-voltage generator in this collection portion may double as the charged electrode and the high-voltage generator described above, or may be disposed separately from these.

[0147] In the method of producing the nonwoven fabric of one or more embodiments, such as described above, the ultrafine fibers can be uniformly and efficiently produced at high speed from the resin mix melt of the component A and the component B', and therefore, the nonwoven fabric of one or more embodiments of the present disclosure can have a larger size can be industrially efficiently produced on an actual produce line. That is, industrial production (e.g., mass production) of the nonwoven fabric of or according to one or more embodiments of the present disclosure can have a wide sheet width with a number average fiber diameter of 4 μm or less is achievable.

[0148] The method of producing the nonwoven fabric of one or more embodiments preferably employs a melting method using a melt. Examples of another means include a solution method that spins using a polymer solution in which a polymer is dissolved in a solvent. The melting method may be preferred because it can allow efficient producing on a producing line since it does not have a

process of solvent recovery or the like and it allows achieving stable production without having concern about a residual solvent and the like in a produced product.

[0149] In the method of producing the fibers of one or more embodiments, as described relating to the nonwoven fabric of one or more embodiments of the present disclosure described above, the fiber diameter of the fiber is preferably 2 μm or less, more preferably 1.5 μm or less, and further preferably 1 μm or less.

[0150] The fiber diameter of the fiber is preferably 0.1 μm or more, and more preferably 0.2 μm or more.

[0151] In the method of producing the nonwoven fabric of one or more embodiments, such as described relating to the nonwoven fabric of one or more embodiments of the present disclosure described above, the number average fiber diameter of the constituent fibers of the nonwoven fabric is preferably 3 μm or less, and more preferably 2 μm or less.

[0152] The number average fiber diameter is preferably 0.1 μm or more, and more preferably 0.2 μm or more.

[0153] In the method of producing the nonwoven fabric of one or more embodiments, as described relating to the nonwoven fabric of one or more embodiments the present disclosure such as described above, the median fiber diameter of the constituent fibers of the nonwoven fabric is preferably 2 μm or less, more preferably 1.5 μm or less, and further preferably 1 μm or less.

[0154] The median fiber diameter is preferably 0.1 μm or more, and more preferably 0.2 μm or more.

EXAMPLES

[0155] Hereinafter, one or more embodiments of the present disclosure will be described more in detail with reference to Examples, but the present disclosure is not limited thereto. Terms “part” and “%” in the Examples are based on mass unless otherwise noted. The symbol “—” in Tables 1 to 3 means the same value as a value in a left column, and the symbol “—” below means that the relevant item has no value.

Examples 1 to 4

[0156] PCL “Capa6250” (trade name, manufactured by Ingevity) as the thermoplastic resin of the component A and stearic acid “LUNAC S-70V” (trade name, manufactured by Kao Corporation) as the compound of the component B were mixed in the proportions shown in Table 1, and thus, a thermoplastic resin composition was made. A melt of the thermoplastic resin composition was fabricated, and spinning was performed under the spinning conditions shown in Table 1 using one nozzle having the hole diameter shown in Table 1. Simultaneously with the spinning, the fibers were collected into a sheet shape, and thus, respective nonwoven fabric samples of Examples 1 to 4 were formed. In Example 1, respective DSC (temperature decrease) results of the component A, the component B, and the thermoplastic resin composition fabricated by mixing the two were as illustrated in FIG. 2.

Example 5

[0157] A nonwoven fabric sample was fabricated similarly to Example 2 except that behenic acid “NAA-222S” (trade name, manufactured by NOF CORPORATION) was used as the compound of the component B.

Example 6

[0158] A nonwoven fabric sample was fabricated similarly to Example 4 except that behenic acid was used as the compound of the component B.

Example 7

[0159] A nonwoven fabric sample was fabricated similarly to Example 2 except that glyceryl behenate “Sunsoft No. 8100-C” (trade name, manufactured by Taiyo Kagaku Co., Ltd.) was used as the compound of the component B.

Example 8

[0160] A nonwoven fabric sample was fabricated similarly to Example 2 except that distearyldimonium chloride “Varisoft TA 100” (trade name, manufactured by Evonik Operations GmbH) was used as the compound of the component B.

Example 9

[0161] A nonwoven fabric sample was fabricated similarly to Example 2 except that N-(hexadecyloxyhydroxypropyl)-N-hydroxyethylhexadecanamide “Sphingolipid E” (trade name, manufactured by Kao Corporation) was used as the compound of the component B.

Example 10

[0162] A nonwoven fabric sample was fabricated similarly to Example 7 except that the negative electric charge shown in Table 2 was applied using sodium stearoyl lactylate (SSL) (manufactured by Musashino Chemical Laboratory, Ltd.) as a charging agent.

Example 11

[0163] A nonwoven fabric sample was fabricated similarly to Example 10 except that PCL with two types of mass average molecular weights as shown in Table 2 was used as the thermoplastic resin of the component A.

Example 12

[0164] A nonwoven fabric sample was fabricated similarly to Example 2 except that glyceryl stearate “EXCEL S-95” (trade name, manufactured by Kao Corporation) was used as the compound of the component B.

Example 13

[0165] A nonwoven fabric sample of Example 13 was fabricated similarly to Example 2 except that stearic acid monoamide “ALFLOW S-10” (trade name, manufactured by NOF CORPORATION) was used as the compound of the component B, and spinning was performed under the spinning conditions shown in Table 2 using one nozzle having the hole diameter shown in Table 2.

Example 14

[0166] A nonwoven fabric sample of Example 14 was fabricated similarly to Example 13 except that myristyl alcohol “KALCOL 4098” (trade name, manufactured by Kao Corporation) was used as the compound of the component B, and the temperature of the heating fluid was set to 130° C.

Example 15

[0167] A nonwoven fabric sample was fabricated similarly to Example 13 except that zinc stearate “ZINC STEARATE G” (trade name, manufactured by NOF CORPORATION) was used as the compound of the component B, and the heating temperature during melt forming and the heating fluid temperature were set to 150° C.

Example 16

[0168] A nonwoven fabric sample was fabricated similarly to Example 15 except that glyceryl behenate was used as the compound of the component B, the additive amount was 5 mass %, and the heating temperature during melt forming was set to 120° C.

Example 17

[0169] A nonwoven fabric sample was fabricated similarly to Example 16 except that the additive amount of the component B was 11 mass %.

Example 18

[0170] A nonwoven fabric sample was fabricated similarly to Example 16 except that the additive amount of the component B was 40 mass %.

Comparative Example 1

[0171] Spinning was performed similarly to Example 1 except that the compound of the component B was not used. However, fiberization failed, and thus, produce of the nonwoven fabric failed.

Comparative Example 2

[0172] A nonwoven fabric sample was fabricated similarly to Example 2 except that lauryl alcohol “KALCOL 2098” (trade name, manufactured by Kao Corporation) was used as an additive of the component B, and the spinning conditions were as shown in Table 3.

Comparative Example 3

[0173] Spinning was attempted similarly to Example 2 except that N-stearoyl-N-methyltaurinesodium “NIKKOL-SMT” (trade name, manufactured by Nikko Chemicals Co., Ltd.) was used as the additive of the component B, the heating temperature during melt forming was set to 180° C., and the spinning conditions were as shown in Table 3, but the nozzle was obstructed, and produce of the nonwoven fabric failed.

Comparative Example 4

[0174] A nonwoven fabric sample was fabricated similarly to Example 2 except that glyceryl behenate was used as the additive of the component B, the additive amount was 1 mass %, and the spinning conditions were as shown in Table 3.

Comparative Example 5

[0175] Spinning was attempted using glyceryl behenate as the additive of the component B and setting the additive amount to 60 mass %, but mixture failed, and thus, spinning failed.

TABLE-US-00001 TABLE 1 Ex. 1 Ex. 2 Ex. 3 Ex. 4 Ex. 5 Ex. 6 Component A Kind — PCL — Thermoplastic Mass average g/mol 25 000 — resin molecular weight (Mw) Content mass % 90 80 70 80 — percentage Solidification ° C. 30 — point Melting point ° C. 60 — Component B Kind — Stearic — Behenic — Compound acid acid Classification — Fatty — acid Molecular g/mol 284.48 — 340.58 — weight (Mw) Content mass % 10 20 30 20 — percentage Solidification ° C. 67 — 74 — point Melting point ° C. 70 — 78 — Charging Kind — None — agent Additive — — — — — amount Ex. 7 Ex. 8 Ex. 9 Component A Kind — PCL — Thermoplastic Mass average g/mol 25 000 — resin molecular weight (Mw) Content mass % 80 — percentage Solidification ° C. 30 — point Melting ° C. 60 — point Component B Kind — Glyceryl Distearoyldimonium N- Compound behenate chloride (hexadecyloxyhydro- xypropyl)-N- hydroxyethylhexa decanamide Classification — Polyhydric Alkyl Sphingolipid alcohol ammonium organic acid salt ester compound Molecular g/mol 414.66 586.50 513.85 weight (Mw) Content mass % 20 — percentage Solidification ° C. 78 69 52 point Melting ° C. 81 71 77 point Charging Kind — None — agent Additive — — — — — amount Ex. 1 Ex. 2 Ex. 3 Ex. 4 Ex. 5 Ex. 6 Ex. 7 Ex. 8 Ex. 9 Spinning Voltage kV None — — — — — condition application Heating ° C. 120 — — — — — temperature during melt forming Heating ° C. 150 — — 190 150 190 — fluid temperature Discharge g/minute 0.5 — 5.0 0.5 5.0 0.5 — speed nozzle Discharge min 0.50 — — 0.05 0.50 0.05 0.50 — time Hole μm 250 — — — — — diameter of nozzle head portion Spinning Fiberization — Good — — — — — result Number μm 1.9 1.4 1.1 1.9 1.4 2.5 1.2 2.0 1.6 average fiber diameter Median μm 0.7 0.9 0.8 1.1 1.0 1.5 0.9 1.1 1.0 fiber diameter Sheet width mm 300 — — — — — Remarks: ‘Ex.’ means Example. Remarks: ‘Ex.’ means Example. Remarks: ‘Ex.’ means Example.

TABLE-US-00002 TABLE 2 Ex. 10 Ex. 11 Ex. 12 Ex. 13 Component A Kind — PCL — Thermoplastic Mass g/mol 25 000 10 000 25 000 — resin average (69 wt %) molecular 50 000 weight (31 wt %) (Mw) Content mass % 79 — 80 — percentage Solidification ° C. 30 — — point Melting ° C. 60 — — point Component B Kind — Glyceryl — Glyceryl Stearic acid Compound behenate stearate monoamide Classification — Polyhydric — Fatty acid alcohol organic amide acid ester compound compound Molecular g/mol 414.66 — 358.56 283.49 weight (Mw) Content mass % 20 — — percentage Solidification ° C. 78 — 66 99 point Melting ° C. 81 — 71 101 point Charging Kind — SSL — None — agent Additive — 1 — — — — — amount Ex. 14 Ex. 15 Ex. 16 Ex. 17 Ex. 18 Component A Kind — PCL — Thermoplastic Mass average g/mol 25 000 — resin molecular weight (Mw) Content mass % 80 — 95 89 60 percentage Solidification ° C. 30 — — — point Melting ° C. 60 — — — point Component B Kind — Myristyl Zinc Glyceryl — Compound alcohol stearate behenate Classification — Higher Metallic Polyhydric — alcohol soap alcohol (fatty organic acid acid ester metal compound salt) Molecular g/mol 270.49 632.33 414.66 — weight (Mw) Content mass % 20 — 5 11 40 percentage Solidification ° C. 36 104 78 — point Melting ° C. 39 124 81 — point Charging Kind — None — — — — agent Additive — — — — — amount Ex. 10 Ex. 11 Ex. 12 Ex. 13 Spinning Voltage application kV —10 — None — condition Heating temperature ° C. 120 — — — during melt forming Heating fluid ° C. 150 — — — temperature Discharge speed g/minute 0.5 — — 0.007 nozzle Discharge time min 0.50 — — 10 Hole diameter of μm 250 — — — nozzle head portion Spinning Fiberization — Good — — — result Number average μm 2.6 1.2 3.2 1.2 fiber diameter Median fiber μm 1.0 0.8 2.3 1.3 diameter Sheet width mm 300 — — 100 Ex. 14 Ex. 15 Ex. 16 Ex. 17 Ex. 18 Spinning Voltage application kV None — — — — condition Heating temperature ° C. 120 150 120 — — during melt forming Heating fluid ° C. 130 150 — — temperature Discharge speed g/minute 0.007 — — — nozzle Discharge time min 10 — — — Hole diameter of μm 250 — — — nozzle head portion Spinning Fiberization — Good — — — result Number average μm 0.4 0.8 0.9 1.1 0.9 fiber diameter Median fiber μm 0.2 — 0.3 0.6 — diameter Sheet width mm 100 — — — Remarks: ‘Ex.’ means Example.

TABLE-US-00003 TABLE 3 CEx. 1 CEx. 2 CEx. 3 Component A Kind — PCL — Thermoplastic Mass average g/mol 25 000 — resin molecular weight (Mw) Content percentage mass % 100 80 — Solidification point ° C. 30 — — Melting point ° C. 60 — — Component B Kind — None Lauryl N-stearoyl- Compound alcohol N-methyltaurine Na Classification — None Higher Acyl alkyl alcohol taurine salt Molecular weight (Mw) g/mol — 186.34 427.60 Content percentage mass % — 20 — Solidification point ° C. — 20 179 Melting point ° C. — 24 193 Charging Kind — None — — agent Additive amount — — — — CEx. 4 CEx. 5

Component A Kind — PCL ← Thermoplastic Mass average g/mol 25 000 ← resin molecular weight (Mw) Content percentage mass % 99.40 Solidification point ° C. 30 ← Melting point ° C. 60 ← Component B Kind — Glyceryl behenate ← Compound Classification — Polyhydric alcohol ← organic acid ester Molecular weight (Mw) g/mol 414.66 414.66 Content percentage mass % 1.60 Solidification point ° C. 78 ← Melting point ° C. 81 ← Charging Kind — None ← agent Additive amount — — ← CEx. 1 CEx. 2 CEx. 3 CEx. 4 CEx. 5 Spinning Voltage application kV None ← ← ← ← condition Heating temperature ° C. 120 ← 180 120 ← during melt forming Heating fluid ° C. 150 ← ← ← — temperature Discharge speed g/minute 0.5 ← ← ← — nozzle Discharge time min — 0.50 — 0.50 — Hole diameter of μm 250 ← ← ← — nozzle head portion Spinning Fiberization — Fusion Good Nozzle Good Mixture result obstruction failure Number average μm — 5.9 — 4.7 — fiber diameter Median fiber μm — 5.7 — 2.6 — diameter Sheet width mm — 300 — 300 — Remarks: 'CEx.' means Comparative Example.

[0176] In Comparative Example 1, solidification during spinning failed, lumps are formed from the melted thermoplastic resin composition as illustrated in FIG. 4, and the fiber was not produced. In contrast to this, in Example 1, as illustrated in FIGS. 3(A) to 3(C), the ultrafine fibers with a median of the fiber diameters of approximately 1 μm were uniformly formed, and as a result, the nonwoven fabric sample having a sheet width of 300 mm with a number average fiber diameter of 1.9 μm and a median fiber diameter of 0.7 μm was able to be produced as shown in Table 1.

[0177] In Examples 2 to 18, the fibers were satisfactorily formed to make the nonwoven fabrics, and thus, the nonwoven fabric samples having the number average fiber diameters, the median fiber diameters, and the sheet widths shown in Tables 1 and 2 were able to be produced.

[0178] Having described our invention as related to this embodiments and Examples, it is our intention that neither the present disclosure nor embodiments thereof be limited by any of the details of the description, unless otherwise specified, but rather be construed broadly within its spirit and scope as set out in the accompanying claims.

[0179] Embodiments of the disclosed subject matter can also be as set forth according to the following parentheticals.

[0180] [1] A method of producing fibers that spins a thermoplastic resin composition including 50 mass % or more and 95 mass % or less of a component A below and 5 mass % or more and 50 mass % or less of a component B' below with respect to a mass of the whole thermoplastic resin composition with a fiber diameter of 4 μm or less, and the method comprising: [0181] a process of heating and melting the thermoplastic resin composition (I); and [0182] a process of discharging the thermoplastic resin composition from a nozzle (II), [0183] wherein the component A is a thermoplastic resin having a solidification point of 100° C. or less, and [0184] the component B' is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than a process temperature.

[0185] [2] The method of producing fibers according to [1], wherein in the process (II), a process of spraying a heating fluid is performed.

[0186] [3] The method of producing fibers according to [1] or [2], [0187] wherein the heating fluid has a temperature higher than the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A, and a difference between the temperature of the heating fluid and the solidification point is preferably 30° C. or more and 200° C. or less, more preferably 40° C. or more and 150° C. or less, and further preferably 50° C. or more and 130° C. or less.

[0188] [4] The method of producing fibers according to any one of [1] to [3], wherein in the process (II), an electrostatic spinning process is performed.

[0189] [5] The method of producing fibers according to any one of [1] to [4], wherein in the process (II), a discharge speed of a melt of the thermoplastic resin composition from the nozzle is 0.1 g/minute-nozzle or more, preferably 0.1 g/minute.Math.nozzle or more and 20 g/minute.Math.nozzle or less, more preferably 0.2 g/minute.Math.nozzle or more and 10 g/minute.Math.nozzle or less, and further preferably 0.5 g/minute.Math.nozzle or more and 10 g/minute.Math.nozzle or less.

[0190] [6] The method of producing fibers according to any one of [1] to [5], wherein the component B' includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyhydric alcohol organic acid ester compounds, waxes, sphingolipids, alkyl ammonium salts, and fatty acid metal salts.

[0191] [7] The method of producing fibers according to any one of [1] to [6], wherein the component A includes a thermoplastic resin having a solidification point of 0° C. or more and 75° C. or less, and preferably 0° C. or more and 60° C. or less.

[0192] [8] The method of producing fibers according to any one of [1] to [7], wherein the process temperature is higher than a melting point of a thermoplastic resin having the highest melting point among the thermoplastic resins, and a difference between the process temperature and the melting point is 20° C. or more and 200° C. or less.

[0193] [9] The method of producing fibers according to any one of [1] to [8], wherein the thermoplastic resin composition includes 60 mass % or more and 95 mass % or less, preferably 65 mass % or more and less than 90 mass %, and more preferably 70 mass % or more and 85 mass % or less of the component A with respect to a whole mass of the thermoplastic resin composition.

[0194] [10] The method of producing fibers according to any one of [1] to [9], wherein the thermoplastic resin composition includes 10 mass % or more and 40 mass % or less, preferably more than 10 mass % and 35 mass % or less, and more preferably more than 10 mass % and 30 mass % or less of the component B' with respect to a whole mass of the thermoplastic resin composition.

[0195] [11] The method of producing fibers according to any one of [1] to [10], wherein the fiber has a fiber diameter of 0.1 μm or more and 2 μm or less, preferably 0.2 μm or more and 1.5 μm or less, and more preferably 0.2 μm or more and 1 μm or less.

[0196] [12] A method of producing a nonwoven fabric, comprising a process of collecting fibers obtained by the method of producing fibers according to any one of [1] to [11], and forming the fibers into a sheet shape.

[0197] [13] The method of producing a nonwoven fabric according to any one of [1] to [12], wherein the nonwoven fabric has a median fiber diameter of the constituent fibers of 0.1 μm or more and 2 μm or less, preferably 0.2 μm or more and 1.5 μm or less, and more preferably 0.2 μm or more and 1 μm or less.

[0198] [14] A nonwoven fabric including fibers, comprising: a component A below; and a component B below, inside the fibers, wherein the nonwoven fabric includes 50 mass % or more and 95 mass % or less of the component A and 5 mass % or more and 50 mass % or less of the component B with respect to a mass of the whole nonwoven fabric, the fibers included in the nonwoven fabric have a number average fiber diameter of 4 μm or less, the nonwoven fabric has a sheet width of 50 mm or more, the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic

resins and a melting point lower than 150° C.

[0199] [15] The nonwoven fabric according to [14], wherein the component B includes a compound having a mass average molecular weight of 1 000 g/mol or less, and preferably includes a compound having a mass average molecular weight of 100 g/mol or more and 1 000 g/mol or less.

[0200] [16] The nonwoven fabric according to [14] or [15], wherein the component B includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyhydric alcohol organic acid ester compounds, waxes, sphingolipids, alkyl ammonium salts, and fatty acid metal salts, and the component B preferably includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyglyceryl fatty acid ester compounds, sucrose fatty acid ester compounds, and sorbitan fatty acid ester compounds.

[0201] [17] The nonwoven fabric according to any one of [14] to [16], wherein a percentage of compounds having a salt structure among the compounds included in the component B is more than 0 mass % and 5 mass % or less, preferably 0.5 mass % or more and 3 mass % or less, and more preferably 0.5 mass % or more and 1 mass % or less.

[0202] [18] The nonwoven fabric according to any one of [14] to [17], wherein the nonwoven fabric has a median fiber diameter of the constituent fibers of 2 μm or less.

[0203] [19] The nonwoven fabric according to any one of [14] to [18], wherein the component A includes a thermoplastic resin having a solidification point of 0° C. or more and 75° C. or less, and preferably 0° C. or more and 60° C. or less.

[0204] [20] The nonwoven fabric according to any one of [14] to [19], wherein the component A includes a thermoplastic resin having a melting point of 50° C. or more and 150° C. or less, preferably 50° C. or more and 120° C. or less, and more preferably 50° C. or more and 90° C. or less.

[0205] [21] The nonwoven fabric according to any one of [14] to [20], wherein the nonwoven fabric includes 60 mass % or more and 95 mass % or less, preferably 65 mass % or more and less than 90 mass %, and more preferably 70 mass % or more and 85 mass % or less of the component A with respect to a mass of the whole nonwoven fabric.

[0206] [22] The nonwoven fabric according to any one of [14] to [21], wherein, when focusing on a combination where a difference between a melting point of a thermoplastic resin included in the component A and a melting point of a compound included in the component B becomes the largest, the difference has a value of preferably more than 0° C. and 100° C. or less, more preferably more than 0° C. and 60° C. or less, and further preferably more than 0° C. and 50° C. or less.

[0207] [23] The nonwoven fabric according to any one of [14] to [22], wherein the component B includes a compound having a melting point of 30° C. or more and less than 150° C., preferably 40° C. or more and 135° C. or less, and more preferably 50° C. or more and 120° C. or less.

[0208] [24] The nonwoven fabric according to any one of [14] to [23], wherein the nonwoven fabric includes 10 mass % or more and 40 mass % or less, preferably more than 10 mass % and 35 mass % or less, and more preferably more than 10 mass % and 30 mass % or less of the component B with respect to the mass of the whole nonwoven fabric.

[0209] [25] The nonwoven fabric according to any one of [14] to [24], wherein the nonwoven fabric has a number average fiber diameter of the constituent fibers of 0.1 μm or more and 3 μm or less, and preferably 0.2 μm or more and 2 μm or less.

[0210] [26] The nonwoven fabric according to any one of [14] to [25], wherein the nonwoven fabric has a median fiber diameter of the constituent fibers of 0.1 μm or more and 2 μm or less, preferably 0.2 μm or more and 1.5 μm or less, and more preferably 0.2 μm or more and 1 μm or less.

[0211] [27] The nonwoven fabric according to any one of [14] to [26], wherein the nonwoven fabric has a sheet width of 50 mm or more and 3 000 mm or less, preferably 100 mm or more and 3 000 mm or less, more preferably 150 mm or more and 3 000 mm or less, further preferably 200 mm or more and 3 000 mm or less, and further more preferably 300 mm or more and 3 000 mm or less.

[0212] [28] The nonwoven fabric according to any one of [14] to [27], wherein the nonwoven fabric has “sheet width”/“length perpendicular to sheet width” of 1 or more and 8 or less, preferably 1 or more and 4 or less, more preferably 1 or more and 2 or less, and further preferably 1 or more and 1.5 or less.

[0213] [29] The nonwoven fabric according to any one of [14] to [28], wherein the component A includes one or two or more selected from polyester resins and polyether resins.

[0214] [30] The nonwoven fabric according to any one of [14] to [29], wherein the component A includes one or two or more selected from polycaprolactone, polybutylene succinate, polybutylene succinate adipate, polydioxanone, and polyglycolic acid.

[0215] [31] The nonwoven fabric according to any one of [14] to [30], wherein the component A includes polycaprolactone.

[0216] [32] The nonwoven fabric according to any one of [14] to [31], wherein the thermoplastic resin included in the component A has a mass average molecular weight of 5 000 g/mol or more and 200 000 g/mol or less, preferably 10 000 g/mol or more and 150 000 g/mol or less, and more preferably 10 000 g/mol or more and 100 000 g/mol or less.

[0217] [33] The nonwoven fabric according to any one of [14] to [32], wherein the component B includes one or two or more selected from stearic acid, behenic acid, myristyl alcohol, and glyceryl behenate, and preferably includes one or two or more selected from stearic acid, behenic acid, and glyceryl behenate.

DESCRIPTION OF SYMBOLS

[0218] **1** Constituent fiber [0219] **2** Core portion layer of component A [0220] **3** Skin layer of component B

Claims

1. A method of producing fibers that spins a thermoplastic resin composition including 50 mass % or more and 95 mass % or less of a component A below and 5 mass % or more and 50 mass % or less of a component B' below with respect to a mass of the whole thermoplastic resin composition with a fiber diameter of 4 μm or less, and the method comprising: a process of heating and melting the thermoplastic resin composition (I); and a process of discharging the thermoplastic resin composition from a nozzle (II), wherein the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B' is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than a process temperature.

2. The method of producing fibers according to claim 1, wherein in the process (II), a process of spraying a heating fluid is performed.
3. The method of producing fibers according to claim 1, wherein the heating fluid has a temperature higher than the solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins included in the component A, and a difference between the temperature of the heating fluid and the solidification point is 30° C. or more and 200° C. or less.
4. The method of producing fibers according to claim 1, wherein in the process (II), an electrostatic spinning process is performed.
5. The method of producing fibers according to claim 1, wherein in the process (II), a discharge speed of a melt of the thermoplastic resin composition from the nozzle is 0.1 g/minute.Math.nozzle or more and 20 g/minute.Math.nozzle or less.
6. The method of producing fibers according to claim 1, wherein the component A includes a thermoplastic resin having a solidification point of 0° C. or more and 75° C. or less.
7. The method of producing fibers according to claim 1, wherein the process temperature is higher than a melting point of a thermoplastic resin having the highest melting point among the thermoplastic resins, and a difference between the process temperature and the melting point is 20° C. or more and 200° C. or less.
8. The method of producing fibers according to claim 1, wherein the thermoplastic resin composition includes either 60 mass % or more and 95 mass % or less of the component A with respect to a whole mass of the thermoplastic resin composition and/or the thermoplastic resin composition includes 10 mass % or more and 40 mass % or less of the component B' with respect to a whole mass of the thermoplastic resin composition.
9. The method of producing fibers according to claim 1, wherein the fiber has a fiber diameter of 0.1 μm or more and 2 μm or less.
10. A method of producing a nonwoven fabric, comprising a process of collecting fibers obtained by the method of producing fibers according to claim 1 and forming the fibers into a sheet shape, wherein the nonwoven fabric has a median fiber diameter of the constituent fibers of 0.1 μm or more and 2 μm or less.
11. A nonwoven fabric including fibers, comprising: a component A below; and a component B below, inside the fibers, wherein the nonwoven fabric includes 50 mass % or more and 95 mass % or less of the component A and 5 mass % or more and 50 mass % or less of the component B with respect to a mass of the whole nonwoven fabric, the fibers included in the nonwoven fabric have a number average fiber diameter of 4 μm or less, the nonwoven fabric has a sheet width of 50 mm or more, the component A is a thermoplastic resin having a solidification point of 100° C. or less, and the component B is a compound having a solidification point higher than a solidification point of the thermoplastic resin having the highest solidification point among the thermoplastic resins and a melting point lower than 150° C.
12. The nonwoven fabric according to claim 11, wherein the component B includes a compound having a mass average molecular weight of 1 000 g/mol or less, and includes a compound having a mass average molecular weight of 100 g/mol or more and 1 000 g/mol or less.
13. The nonwoven fabric according to claim 11, wherein the component B includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyhydric alcohol organic acid ester compounds, waxes, sphingolipids, alkyl ammonium salts, and fatty acid metal salts, and the component B includes one or two or more selected from fatty acids, higher alcohols, fatty acid amide compounds, polyglyceryl fatty acid ester compounds, sucrose fatty acid ester compounds, and sorbitan fatty acid ester compounds.
14. The nonwoven fabric according to claim 11, wherein the nonwoven fabric has a median fiber diameter of the constituent fibers of 2 μm or less.
15. The nonwoven fabric according to claim 11, wherein the component A includes a thermoplastic resin having a solidification point of 0° C. or more and 75° C. or less, and/or the component A includes a thermoplastic resin having a melting point of 50° C. or more and 150° C. or less.
16. The nonwoven fabric according to claim 11, wherein the nonwoven fabric includes 60 mass % or more and 95 mass % or less of the component A with respect to a mass of the whole nonwoven fabric.
17. The nonwoven fabric according to claim 11, wherein the nonwoven fabric includes 10 mass % or more and 40 mass % or less of the component B with respect to the mass of the whole nonwoven fabric.
18. The nonwoven fabric according to claim 11, wherein the nonwoven fabric has a number average fiber diameter of the constituent fibers of 0.1 μm or more and 3 μm or less, and/or the nonwoven fabric has a median fiber diameter of the constituent fibers of 0.1 μm or more and 2 μm or less.
19. The nonwoven fabric according to claim 11, wherein the nonwoven fabric has a sheet width of 50 mm or more and 3 000 mm or less and/or the nonwoven fabric has “sheet width”/“length perpendicular to sheet width” of 1 or more and 8 or less.
20. The nonwoven fabric according to claim 11, wherein the component A includes one or two or more selected from polycaprolactone, polybutylene succinate, polybutylene succinate adipate, polydioxanone, and polyglycolic acid, and wherein the component A includes polycaprolactone.
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